WORKED SOLUTION

STEP III 1996, Question 1

Pure • Powers of Hyperbolic Functions

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s3-q1

The key idea

Everything follows from writing hyperbolic functions as exponentials. A power $\cosh^n x$ becomes $2^{-n}(e^x + e^{-x})^n$, which the binomial theorem expands into exponentials $e^{(n-2r)x}$; these come in matching pairs $e^{\pm jx}$, and each pair is a single $\cosh jx$. The same expansion for $\sinh^{2m} x$ differs only by alternating signs, so adding or subtracting kills every other term.

Solution

Definitions and the two identities

We define

$$\cosh x = \frac{e^x + e^{-x}}{2}, \qquad \sinh x = \frac{e^x - e^{-x}}{2}.$$

Two facts follow at once. Since $\cosh x + \sinh x = e^x$ and $\cosh x - \sinh x = e^{-x}$,

$$\cosh^2 x - \sinh^2 x = (\cosh x + \sinh x)(\cosh x - \sinh x) = e^x e^{-x} = 1.$$

Squaring the definitions, $\cosh^2 x = \frac{1}{4}(e^{2x} + 2 + e^{-2x})$ and $\sinh^2 x = \frac{1}{4}(e^{2x} - 2 + e^{-2x})$, so

$$\cosh^2 x + \sinh^2 x = \tfrac{1}{2}(e^{2x} + e^{-2x}) = \cosh 2x,$$

and, adding or subtracting the identity $\cosh^2 x - \sinh^2 x = 1$,

$$\cosh^2 x = \tfrac{1}{2}(\cosh 2x + 1), \qquad \sinh^2 x = \tfrac{1}{2}(\cosh 2x - 1).$$

The last two hold for any argument, so we may replace x by $2x$ in them.

Now factorise the difference of squares:

$$\cosh^4 x - \sinh^4 x = (\cosh^2 x - \sinh^2 x)(\cosh^2 x + \sinh^2 x) = 1 \cdot \cosh 2x = \cosh 2x,$$

as required. For the sum, complete the square and use $\cosh x \sinh x = \frac{1}{4}(e^{2x} - e^{-2x}) = \frac{1}{2}\sinh 2x$:

$$\begin{aligned}\cosh^4 x + \sinh^4 x &= (\cosh^2 x + \sinh^2 x)^2 - 2\cosh^2 x \sinh^2 x \\ &= \cosh^2 2x - \tfrac{1}{2}\sinh^2 2x \\ &= \tfrac{1}{2}(\cosh 4x + 1) - \tfrac{1}{4}(\cosh 4x - 1) \\ &= \tfrac{1}{4}\cosh 4x + \tfrac{3}{4},\end{aligned}$$

which is the second identity.

The coefficients a_k

By the binomial theorem,

$$\cosh^n x = \frac{1}{2^n}\left(e^x + e^{-x}\right)^n = \frac{1}{2^n}\sum_{r=0}^{n}\binom{n}{r}e^{(n-2r)x}.$$

Pair the term with index r with the term with index $n-r$. They have the same coefficient, since $\binom{n}{r} = \binom{n}{n-r}$, and opposite exponents $\pm(n-2r)x$, so together they give $2\binom{n}{r}\cosh((n-2r)x)$.

> **Ask yourself** Which multiples $\cosh kx$ actually appear, and what happens to the middle term?

The exponents $n-2r$ are $n, n-2, n-4, \ldots$, so only k with the same parity as n occur. Writing $k = n-2r$, the coefficient of $\cosh kx$ for $1 \le k \le n$ is $2\cdot 2^{-n}\binom{n}{r}$ with $r = \frac{1}{2}(n-k)$. When n is even there is also an unpaired middle term $r = \frac{n}{2}$, giving the constant $2^{-n}\binom{n}{n/2}$. Hence

$$a_k = \begin{cases} \dfrac{1}{2^{n-1}}\dbinom{n}{\frac{1}{2}(n-k)} & \text{if } 1 \le k \le n \text{ and } n-k \text{ is even,} \\ 0 & \text{if } n-k \text{ is odd,}\end{cases} \qquad a_0 = \begin{cases} \dfrac{1}{2^n}\dbinom{n}{\frac{1}{2}n} & n \text{ even,} \\ 0 & n \text{ odd.}\end{cases}$$

For example, $\cosh^5 x = \frac{5}{8}\cosh x + \frac{5}{16}\cosh 3x + \frac{1}{16}\cosh 5x$.

The final part

Take $n = 2m$ and expand $\sinh^{2m} x$ in the same way:

$$\sinh^{2m} x = \frac{1}{2^{2m}}\sum_{r=0}^{2m}(-1)^r\binom{2m}{r}e^{(2m-2r)x}.$$

The terms r and $2m-r$ still pair up, because $(-1)^{2m-r} = (-1)^r$, so

$$\sinh^{2m} x = \frac{1}{2^{2m}}\sum_{r=0}^{2m}(-1)^r\binom{2m}{r}\cosh(2(m-r)x),$$

which is the expansion of $\cosh^{2m} x$ (written the same way, with cosh even) with alternating signs. Subtracting and adding,

$$\cosh^{2m} x \mp \sinh^{2m} x = \frac{1}{2^{2m}} \sum_{r=0}^{2m} (1 \mp (-1)^r) \binom{2m}{r} \cosh(2(m-r)x).$$

So the difference keeps only odd r (each doubled) and the sum keeps only even r. Collecting the pairs r, $2m-r$ as before, and writing $k = 2j = 2(m-r)$, we get

$$\cosh^{2m} x - \sinh^{2m} x = b_0 + \sum_{\substack{1 \le j \le m \\ m-j \text{ odd}}} \frac{1}{2^{2m-2}} \binom{2m}{m-j} \cosh 2jx,$$

$$\cosh^{2m} x + \sinh^{2m} x = c_0 + \sum_{\substack{1 \le j \le m \\ m-j \text{ even}}} \frac{1}{2^{2m-2}} \binom{2m}{m-j} \cosh 2jx,$$

where the constant terms come from the unpaired middle term $r = m$, which survives in the difference when m is odd and in the sum when m is even:

$$b_0 = \begin{cases} \dfrac{1}{2^{2m-1}} \dbinom{2m}{m} & m \text{ odd}, \\ 0 & m \text{ even}, \end{cases} \qquad c_0 = \begin{cases} \dfrac{1}{2^{2m-1}} \dbinom{2m}{m} & m \text{ even}, \\ 0 & m \text{ odd}. \end{cases}$$

In words: the coefficient of $\cosh kx$ is 0 for odd k. For even $k = 2j \ge 2$ it is $2^{2-2m}\binom{2m}{m-j}$ in the difference when $m-j$ is odd, and in the sum when $m-j$ is even, and 0 otherwise. With $m = 2$ the difference is $\frac{1}{4}\binom{4}{1}\cosh 2x = \cosh 2x$ and the sum is $\frac{1}{8}\binom{4}{2} + \frac{1}{4}\binom{4}{0}\cosh 4x = \frac{3}{4} + \frac{1}{4}\cosh 4x$, recovering the identities we proved first.

Checks

Put $x = 0$: every $\cosh kx$ becomes 1, so the a_k must sum to 1 (they do: the binomial coefficients sum to 2^n), and both final expansions must also sum to 1. For $m = 3$ the formulas give $\cosh^6 x - \sinh^6 x = \frac{5}{8} + \frac{3}{8}\cosh 4x$ and $\cosh^6 x + \sinh^6 x = \frac{15}{16}\cosh 2x + \frac{1}{16}\cosh 6x$. At $x = 1$ both sides of the first are 10.87 to four significant figures, which a calculator confirms. We checked all the formulas numerically for $n \le 12$ and $m \le 8$ at random values of x.

Common slips

- Doubling the middle term: when n is even, the $r = \frac{n}{2}$ term has no partner, so a_0 is $2^{-n}\binom{n}{n/2}$, not twice that.
- Forgetting that $a_k = 0$ whenever k and n have opposite parity.
- Assuming the constant term is always present in the final part. It appears in the difference only for odd m and in the sum only for even m.

Beyond STEP

Writing powers of $\cosh x$ (or $\cos x$) as combinations of $\cosh kx$ is a first taste of Fourier analysis: e^{ikx} are the natural building blocks, and $\cos^n x$ is a finite Fourier series whose coefficients are binomial. Reversing the process, expressing $\cos nx$ as a polynomial in $\cos x$, gives the Chebyshev polynomials, which matter in approximation theory. Needham's *Visual Complex Analysis* shows why the exponential form makes all these identities natural.

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